

# Отчёт по лабораторной работе 5

Простые сети в GNS3. Анализ трафика

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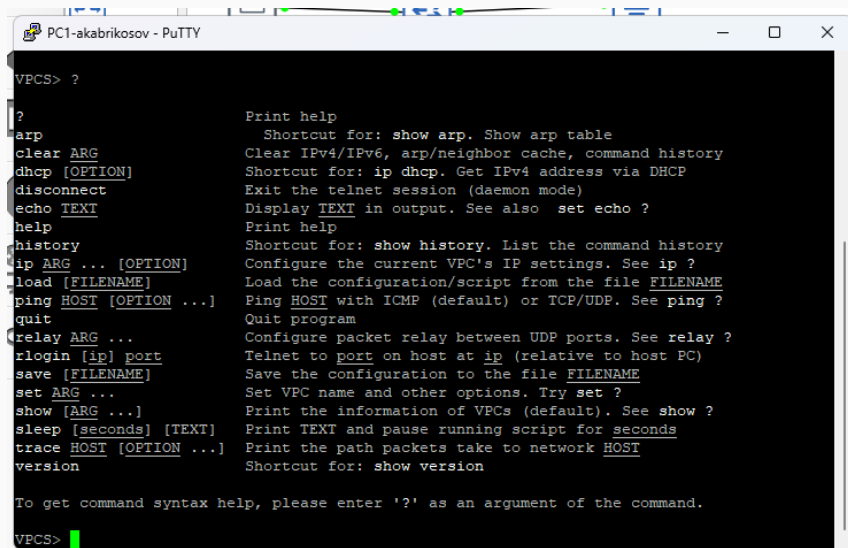
## Цели и задачи работы

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Построение простейших моделей сети на базе коммутатора и маршрутизаторов FRR и VyOS в GNS3, анализ трафика посредством Wireshark.

## Выполнение лабораторной работы

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```
PC1-akabrikosov - PuTTY
VPCS> ?
?                Print help
arp              Shortcut for: show arp. Show arp table
clear ARG       Clear IPv4/IPv6, arp/neighbor cache, command history
dhcp [OPTION]   Shortcut for: ip dhcp. Get IPv4 address via DHCP
disconnect      Exit the telnet session (daemon mode)
echo TEXT       Display TEXT in output. See also set echo ?
help            Print help
history         Shortcut for: show history. List the command history
ip ARG ... [OPTION] Configure the current VPC's IP settings. See ip ?
load [FILENAME] Load the configuration/script from the file FILENAME
ping HOST [OPTION ...] Ping HOST with ICMP (default) or TCP/UDP. See ping ?
quit           Quit program
relay ARG ...   Configure packet relay between UDP ports. See relay ?
rlogin [ip] port Telnet to port on host at ip (relative to host PC)
save [FILENAME] Save the configuration to the file FILENAME
set ARG ...     Set VPC name and other options. Try set ?
show [ARG ...]  Print the information of VPCs (default). See show ?
sleep [seconds] [TEXT] Print TEXT and pause running script for seconds
trace HOST [OPTION ...] Print the path packets take to network HOST
version         Shortcut for: show version

To get command syntax help, please enter '?' as an argument of the command.
VPCS>
```

Рис. 1: Просмотр списка команд VPCS

```
VPCS> ip ?

ip ARG ... [OPTION]
  Configure the current VPC's IP settings
  ARG ...:
    address [mask] [gateway]
    address [gateway] [mask]
    Set the VPC's ip, default gateway ip and network mask
    Default IPv4 mask is /24, IPv6 is /64. Example:
    ip 10.1.1.70/26 10.1.1.65 set the VPC's ip to 10.1.1.70,
    the gateway to 10.1.1.65, the netmask to 255.255.255.192.
    In tap mode, the ip of the tapx is the maximum host ID
    of the subnet. In the example above the tapx ip would be
    10.1.1.126
    mask may be written as /26, 26 or 255.255.255.192
  auto
    Attempt to obtain IPv6 address, mask and gateway using SLAAC
  dhcp [OPTION]
    Attempt to obtain IPv4 address, mask, gateway, DNS via DHCP
    -d      Show DHCP packet decode
    -r      Renew DHCP lease
    -x      Release DHCP lease
  dns ip    Set DNS server ip, delete if ip is '0'
  dns6 ipv6 Set DNS server ipv6, delete if ipv6 is '0'
  domain NAME Set local domain name to NAME

VPCS>
```

Рис. 2: Просмотр синтаксиса команды ip

## Настройка IP-адреса PC1

```
PC1-akabrikosov - PuTTY
-r      Renew DHCP lease
-x      Release DHCP lease
dns ip   Set DNS server ip, delete if ip is '0'
dns6 ipv6 Set DNS server ipv6, delete if ipv6 is '0'
domain NAME Set local domain name to NAME

VPCS>
VPCS> ip 192.168.1.11/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.11 255.255.255.0 gateway 192.168.1.1

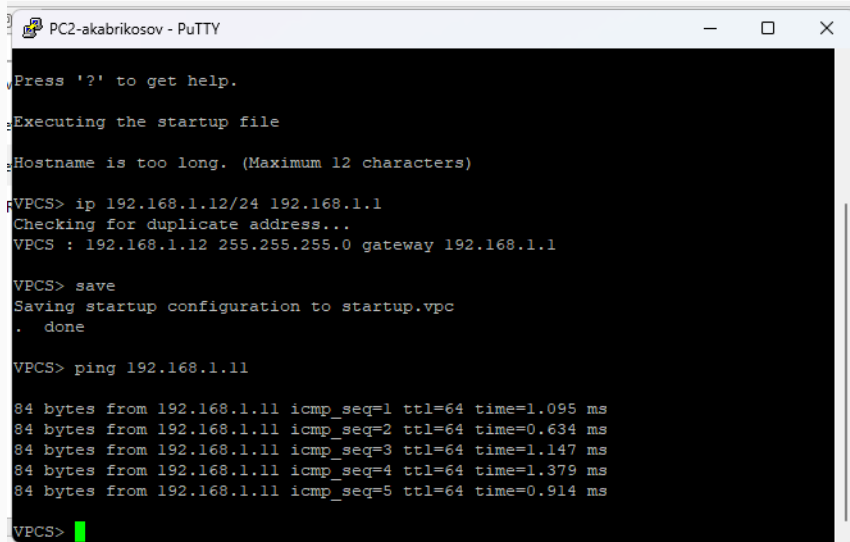
VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.1.11/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10004
RHOST:PORT : 127.0.0.1:10005
MTU        : 1500

VPCS> 
```

## Проверка связности PC1 ↔ PC2



```
PC2-akabrikosov - PuTTY
Press '?' to get help.
Executing the startup file
Hostname is too long. (Maximum 12 characters)
VPCS> ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> ping 192.168.1.11

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=1.095 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=0.634 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.147 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=1.379 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=0.914 ms

VPCS>
```



# ARP-сообщения (gratuitous ARP)

Захват с Standard input [PC1-akabrikosov Ethernet0 to msk-akabrikosov-sw-01 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

| No. | Time     | Source           | Destination | Protocol | Length | Info                                      |
|-----|----------|------------------|-------------|----------|--------|-------------------------------------------|
| 2   | 0.001392 | ::               | ff02::2     | ICMPv6   | 62     | Router Solicitation                       |
| 3   | 0.050615 | Private_66:68:00 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.11 (Request) |
| 4   | 0.052498 | Private_66:68:01 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request) |
| 5   | 1.051895 | Private_66:68:00 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.11 (Request) |
| 6   | 1.053536 | Private_66:68:01 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request) |
| 7   | 2.052518 | Private_66:68:00 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.11 (Request) |
| 8   | 2.053688 | Private_66:68:01 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request) |

> Frame 3: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface ~, id 0

> Ethernet II, Src: Private\_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

▼ Address Resolution Protocol (request/gratuitous ARP)

- Hardware type: Ethernet (1)
- Protocol type: IPv4 (0x0800)
- Hardware size: 6
- Protocol size: 4
- Opcode: request (1)
- [Is gratuitous: True]
- Sender MAC address: Private\_66:68:00 (00:50:79:66:68:00)
- Sender IP address: 192.168.1.11
- Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
- Target IP address: 192.168.1.11

|      |       |
|------|-------|
| 0000 | ff ff |
| 0010 | 08 00 |
| 0020 | ff ff |
| 0030 | 00 00 |

## Опции ping в VPCS

```
PC2-akabrikosov - PuTTY
Checking for duplicate address...
VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

VPCS> ping ?

ping HOST [OPTION ...]
  Ping the network HOST. HOST can be an ip address or name
  Options:
    -1                ICMP mode, default
    -2                UDP mode
    -3                TCP mode
    -c count        Packet count, default 5
    -D                Set the Don't Fragment bit
    -f FLAG         Tcp header FLAG |C|E|U|A|P|R|S|F|
                      bits |7 6 5 4 3 2 1 0|
    -i ms           Wait ms milliseconds between sending each packet
    -l size         Data size
    -P protocol     Use IP protocol in ping packets
                      1 - ICMP (default), 17 - UDP, 6 - TCP
    -p port         Destination port
    -s port         Source port
    -T ttl          Set ttl, default 64
    -t                Send packets until interrupted by Ctrl+C
    -w ms          Wait ms milliseconds to receive the response

  Notes: 1. Using names requires DNS to be set.
         2. Use Ctrl+C to stop the command.

VPCS>
```

```
VPCS> ping 192.168.1.11 -c 1

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=1.726 ms

VPCS> ping 192.168.1.11 -c 1 -2

84 bytes from 192.168.1.11 udp_seq=1 ttl=64 time=1.047 ms

VPCS> ping 192.168.1.11 -c 1 -3

Connect 7@192.168.1.11 seq=1 ttl=64 time=2.851 ms
SendData 7@192.168.1.11 seq=1 ttl=64 time=2.378 ms
Close 7@192.168.1.11 seq=1 ttl=64 time=2.745 ms

VPCS> 
```

Рис. 7: Выполнение ping

## ICMP Echo Request/Reply

|   |    |           |              |              |      |                                                 |
|---|----|-----------|--------------|--------------|------|-------------------------------------------------|
| → | 11 | 80.496250 | 192.168.1.12 | 192.168.1.11 | ICMP | 98 Echo (ping) request id=0x9b0d, seq=1/256, t  |
| ← | 12 | 80.497216 | 192.168.1.11 | 192.168.1.12 | ICMP | 98 Echo (ping) reply id=0x9b0d, seq=1/256, t    |
|   | 13 | 85.157054 | 192.168.1.12 | 192.168.1.11 | ECHO | 98 Request                                      |
|   | 14 | 85.157594 | 192.168.1.11 | 192.168.1.12 | ECHO | 98 Response                                     |
|   | 15 | 90.709579 | 192.168.1.12 | 192.168.1.11 | TCP  | 74 22946 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=146 |
|   | 16 | 90.710591 | 192.168.1.11 | 192.168.1.12 | TCP  | 54 7 → 22946 [SYN, ACK] Seq=0 Ack=1 Win=2920 Le |
|   | 17 | 90.712203 | 192.168.1.12 | 192.168.1.11 | TCP  | 66 22946 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 T |
|   | 18 | 90.713774 | 192.168.1.12 | 192.168.1.11 | ECHO | 122 Request                                     |

|                                               |      |       |
|-----------------------------------------------|------|-------|
| Total Length: 84                              | 0000 | 00 5i |
| Identification: 0x0d9b (3483)                 | 0010 | 00 5i |
| > 000. .... = Flags: 0x0                      | 0020 | 01 0l |
| ...0 0000 0000 0000 = Fragment Offset: 0      | 0030 | 0e 0' |
| Time to Live: 64                              | 0040 | 1e 1' |
| Protocol: ICMP (1)                            | 0050 | 2e 2' |
| Header Checksum: 0xe9a6 [validation disabled] | 0060 | 3e 3' |
| [Header checksum status: Unverified]          |      |       |
| Source Address: 192.168.1.12                  |      |       |
| Destination Address: 192.168.1.11             |      |       |
| [Stream index: 0]                             |      |       |
| Internet Control Message Protocol             |      |       |
| Type: Echo (ping) request (8)                 |      |       |
| Code: 0                                       |      |       |
| Checksum: 0x84fd [correct]                    |      |       |
| [Checksum Status: Good]                       |      |       |
| Identifier (BE): 39693 (0x9b0d)               |      |       |
| Identifier (LE): 3483 (0x0d9b)                |      |       |
| Sequence Number (BE): 1 (0x0001)              |      |       |
| Sequence Number (LE): 256 (0x0100)            |      |       |
| [Response frame: 12]                          |      |       |
| > Data (56 bytes)                             |      |       |

Рис. 8: Анализ ICMP Echo-запроса и ответа

## UDP mode (Echo over UDP)

| No. | Time      | Source       | Destination  | Protocol | Length | Info                                         |
|-----|-----------|--------------|--------------|----------|--------|----------------------------------------------|
| 11  | 80.496250 | 192.168.1.12 | 192.168.1.11 | ICMP     | 98     | Echo (ping) request id=0x9b0d, seq=1/256, t  |
| 12  | 80.497216 | 192.168.1.11 | 192.168.1.12 | ICMP     | 98     | Echo (ping) reply id=0x9b0d, seq=1/256, t    |
| 13  | 85.157054 | 192.168.1.12 | 192.168.1.11 | ECHO     | 98     | Request                                      |
| 14  | 85.157594 | 192.168.1.11 | 192.168.1.12 | ECHO     | 98     | Response                                     |
| 15  | 90.709579 | 192.168.1.12 | 192.168.1.11 | TCP      | 74     | 22946 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=146 |
| 16  | 90.710591 | 192.168.1.11 | 192.168.1.12 | TCP      | 54     | 7 → 22946 [SYN, ACK] Seq=0 Ack=1 Win=2920 Le |
| 17  | 90.712203 | 192.168.1.12 | 192.168.1.11 | TCP      | 66     | 22946 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 T |
| 18  | 90.713774 | 192.168.1.12 | 192.168.1.11 | ECHO     | 122    | Request                                      |

|                                                                                                          |            |
|----------------------------------------------------------------------------------------------------------|------------|
| Identification: 0x0da0 (3488)                                                                            | 0000 00 50 |
| > 000. .... = Flags: 0x0                                                                                 | 0010 00 54 |
| ...0 0000 0000 0000 = Fragment Offset: 0                                                                 | 0020 01 0b |
| Time to Live: 64                                                                                         | 0030 0e 0f |
| Protocol: UDP (17)                                                                                       | 0040 1e 1f |
| Header Checksum: 0xe991 [validation disabled]                                                            | 0050 2e 2f |
| [Header checksum status: Unverified]                                                                     | 0060 3e 3f |
| Source Address: 192.168.1.12                                                                             |            |
| Destination Address: 192.168.1.11                                                                        |            |
| [Stream index: 0]                                                                                        |            |
| ▼ User Datagram Protocol, Src Port: 64052, Dst Port: 7                                                   |            |
| Source Port: 64052                                                                                       |            |
| Destination Port: 7                                                                                      |            |
| Length: 64                                                                                               |            |
| Checksum: 0xe63f [unverified]                                                                            |            |
| [Checksum Status: Unverified]                                                                            |            |
| [Stream index: 0]                                                                                        |            |
| [Stream Packet Number: 1]                                                                                |            |
| > [Timestamps]                                                                                           |            |
| UDP payload (56 bytes)                                                                                   |            |
| ▼ Echo                                                                                                   |            |
| Echo data: 0050796668010e0f101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f3031323334... |            |

## TCP mode (Echo over TCP)

| No. | Time      | Source       | Destination  | Protocol | Length | Info                                         |
|-----|-----------|--------------|--------------|----------|--------|----------------------------------------------|
| 17  | 90.712203 | 192.168.1.12 | 192.168.1.11 | TCP      | 66     | 22946 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 T |
| 18  | 90.713774 | 192.168.1.12 | 192.168.1.11 | ECHO     | 122    | Request                                      |
| 19  | 90.714404 | 192.168.1.11 | 192.168.1.12 | TCP      | 54     | 7 → 22946 [ACK] Seq=1 Ack=57 Win=2920 Len=0  |
| 20  | 90.717374 | 192.168.1.12 | 192.168.1.11 | TCP      | 66     | 22946 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2 |
| 21  | 90.717648 | 192.168.1.11 | 192.168.1.12 | TCP      | 54     | 7 → 22946 [ACK] Seq=1 Ack=58 Win=2920 Len=0  |
| 22  | 90.717693 | 192.168.1.11 | 192.168.1.12 | TCP      | 54     | 7 → 22946 [FIN, ACK] Seq=1 Ack=58 Win=2920 L |
| 23  | 90.720212 | 192.168.1.12 | 192.168.1.11 | TCP      | 66     | 22946 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0  |

|                                                                                      |      |       |
|--------------------------------------------------------------------------------------|------|-------|
| Destination Address: 192.168.1.11                                                    | 0000 | 00 5i |
| [Stream index: 0]                                                                    | 0010 | 00 6i |
| Transmission Control Protocol, Src Port: 22946, Dst Port: 7, Seq: 1, Ack: 1, Len: 56 | 0020 | 01 0i |
| Source Port: 22946                                                                   | 0030 | 0b 6i |
| Destination Port: 7                                                                  | 0040 | 00 0i |
| [Stream index: 0]                                                                    | 0050 | 0d 0i |
| [Stream Packet Number: 4]                                                            | 0060 | 0d 0i |
| [Conversation completeness: Complete, WITH_DATA (31)]                                | 0070 | 0d 0i |
| [TCP Segment Len: 56]                                                                |      |       |
| Sequence Number: 1 (relative sequence number)                                        |      |       |
| Sequence Number (raw): 1822878072                                                    |      |       |
| [Next Sequence Number: 57 (relative sequence number)]                                |      |       |
| Acknowledgment Number: 1 (relative ack number)                                       |      |       |
| Acknowledgment number (raw): 548211389                                               |      |       |
| 1000 .... = Header Length: 32 bytes (8)                                              |      |       |
| Flags: 0x018 (PSH, ACK)                                                              |      |       |
| Window: 2920                                                                         |      |       |
| [Calculated window size: 2920]                                                       |      |       |
| [Window size scaling factor: -2 (no window scaling used)]                            |      |       |
| Checksum: 0x2856 [unverified]                                                        |      |       |
| [Checksum Status: Unverified]                                                        |      |       |
| Urgent Pointer: 0                                                                    |      |       |

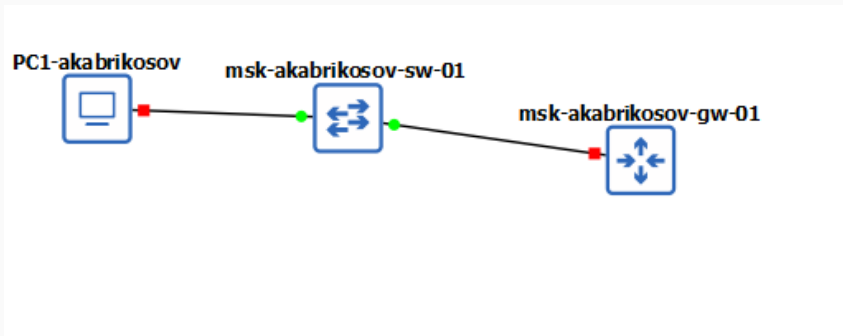


Рис. 11: Топология сети с маршрутизатором FRR

## Настройка IP на PC1 (192.168.1.10/24)

```
VPCS> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 10004
RHOST:PORT  : 127.0.0.1:10005
MTU         : 1500

VPCS> 
```

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## Настройка eth0 на FRR (192.168.1.1/24)

```
msk-akabrikosov-gw-01 - PuTTY

Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

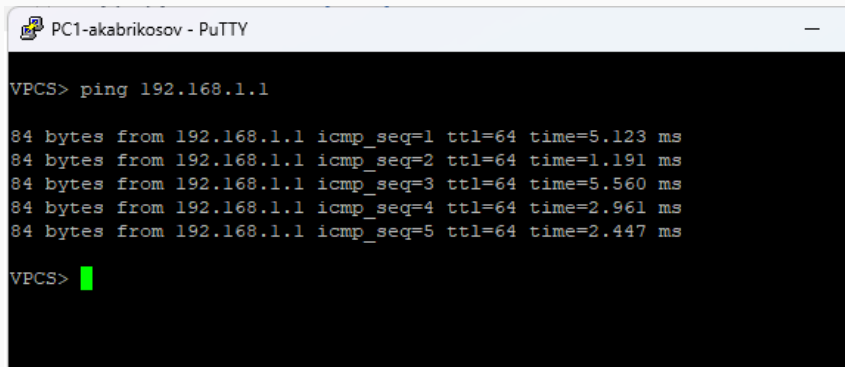
frr# configure terminal
frr(config)# hostname msk-akabrikosov-gw-01
msk-akabrikosov-gw-01(config)# exit
msk-akabrikosov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-akabrikosov-gw-01# configure terminal
msk-akabrikosov-gw-01(config)# interface eth0
msk-akabrikosov-gw-01(config-if)# ip address 192.168.1.1/24
msk-akabrikosov-gw-01(config-if)# no shutdown
msk-akabrikosov-gw-01(config-if)# exit
msk-akabrikosov-gw-01(config)# exit
msk-akabrikosov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-akabrikosov-gw-01#
```

## Проверка конфигурации FRR

```
msk-akabrikosov-gw-01 - PuTTY
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-akabrikosov-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 192.168.1.1/24
exit
!
end
msk-akabrikosov-gw-01# show interface brief
Interface      Status VRF      Addresses
-----
eth0           up     default  192.168.1.1/24
eth1           down   default
eth2           down   default
eth3           down   default
eth4           down   default
eth5           down   default
eth6           down   default
eth7           down   default
slo            up     default
pimreg         up     default
msk-akabrikosov-gw-01#
```

## Проверка связности PC1 → FRR (192.168.1.1)



The image shows a PuTTY terminal window titled "PC1-akabrikosov - PuTTY". The terminal displays the command "VPCS> ping 192.168.1.1" and its output, which consists of five successful ping responses. Each response line shows "84 bytes from 192.168.1.1 icmp\_seq=X ttl=64 time=Y ms", where X is the sequence number (1 to 5) and Y is the round-trip time in milliseconds. The times are 5.123, 1.191, 5.560, 2.961, and 2.447 ms respectively. The prompt "VPCS>" is followed by a green cursor.

```
PC1-akabrikosov - PuTTY

VPCS> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=5.123 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=1.191 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=5.560 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=2.961 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=2.447 ms

VPCS> █
```

Рис. 15: Проверка связи между PC1 и маршрутизатором

# ICMP/ARP в Wireshark (PC1 ↔ FRR)

| No.  | Time       | Source            | Destination      | Protocol | Length | Info                                         |
|------|------------|-------------------|------------------|----------|--------|----------------------------------------------|
| 16   | 211.518461 | 0c:7f:02:2d:00:00 | Private_66:68:00 | ARP      | 60     | 192.168.1.1 is at 0c:7f:02:2d:00:00          |
| 17   | 211.519436 | 192.168.1.10      | 192.168.1.1      | ICMP     | 98     | Echo (ping) request id=0x870f, seq=1/256, t  |
| 18   | 211.523686 | 192.168.1.1       | 192.168.1.10     | ICMP     | 98     | Echo (ping) reply id=0x870f, seq=1/256, t    |
| → 19 | 212.525373 | 192.168.1.10      | 192.168.1.1      | ICMP     | 98     | Echo (ping) request id=0x880f, seq=2/512, t  |
| ← 20 | 212.526083 | 192.168.1.1       | 192.168.1.10     | ICMP     | 98     | Echo (ping) reply id=0x880f, seq=2/512, t    |
| 21   | 213.528184 | 192.168.1.10      | 192.168.1.1      | ICMP     | 98     | Echo (ping) request id=0x890f, seq=3/768, t  |
| 22   | 213.533129 | 192.168.1.1       | 192.168.1.10     | ICMP     | 98     | Echo (ping) reply id=0x890f, seq=3/768, t    |
| 23   | 214.549558 | 192.168.1.10      | 192.168.1.1      | ICMP     | 98     | Echo (ping) request id=0x8a0f, seq=4/1024, t |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>&gt; Frame 19: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0</li> <li>&gt; Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:7f:02:2d:00:00 (0c:7f:02:2d:00:00)</li> <li>▼ Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.1 <ul style="list-style-type: none"> <li>0100 .... = Version: 4</li> <li>.... 0101 = Header Length: 20 bytes (5)</li> <li>&gt; Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT) <ul style="list-style-type: none"> <li>Total Length: 84</li> <li>Identification: 0x0f88 (3976)</li> </ul> </li> <li>&gt; 000. .... = Flags: 0x00 <ul style="list-style-type: none"> <li>...0 0000 0000 0000 = Fragment Offset: 0</li> <li>Time to Live: 64</li> <li>Protocol: ICMP (1)</li> <li>Header Checksum: 0xe7c5 [validation disabled]</li> <li>[Header checksum status: Unverified]</li> <li>Source Address: 192.168.1.10</li> <li>Destination Address: 192.168.1.1</li> <li>[Stream index: 0]</li> </ul> </li> <li>▼ Internet Control Message Protocol <ul style="list-style-type: none"> <li>Type: Echo (ping) request (8)</li> <li>Code: 0</li> <li>Checksum: 0x97fa [correct]</li> <li>[Checksum Status: Good]</li> <li>Identifier (BE): 34831 (0x880f)</li> </ul> </li> </ul> </li> </ul> | <pre> 0000 0c 7f 0010 00 54 0020 01 01 0030 0e 0f 0040 1e 1f 0050 2e 2f 0060 3e 3f </pre> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|

## Моделирование сети на базе VyOS

---

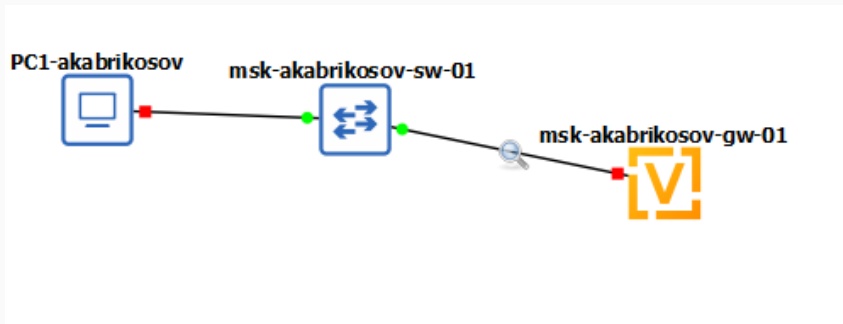
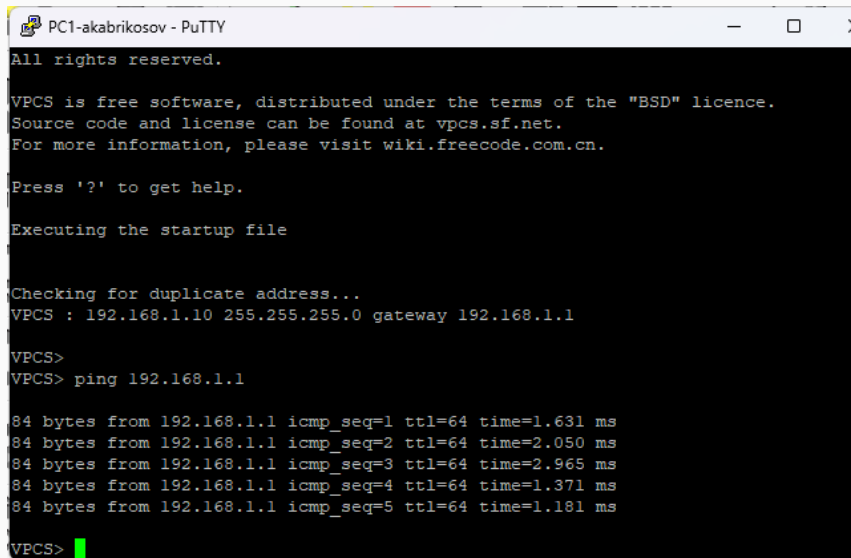


Рис. 17: Топология сети с маршрутизатором VyOS

## Настройка интерфейса eth0 на VyOS (192.168.1.1/24)

```
msk-akabrikosov-gw-01 - PuTTY
[edit]
vyos@vyos# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# show interfaces
  ethernet eth0 {
    address 192.168.1.1/24
    hw-id 0c:8c:c6:b4:00:00
  }
  ethernet eth1 {
    hw-id 0c:8c:c6:b4:00:01
  }
  ethernet eth2 {
    hw-id 0c:8c:c6:b4:00:02
  }
  loopback lo {
  }
[edit]
vyos@vyos#
```

## Проверка связности PC1 → VyOS (192.168.1.1)



```
PC1-akabrikosov - PuTTY
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS>
VPCS> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=1.631 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=2.050 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=2.965 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.371 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.181 ms

VPCS>
```



## ICMP/ARP в Wireshark (PC1 ↔ VyOS)

| No.  | Time       | Source       | Destination  | Protocol | Length | Info                                         |
|------|------------|--------------|--------------|----------|--------|----------------------------------------------|
| 22   | 201.130678 | 192.168.1.10 | 192.168.1.1  | ICMP     | 98     | Echo (ping) request id=0xf11, seq=2/512, t   |
| 23   | 201.132042 | 192.168.1.1  | 192.168.1.10 | ICMP     | 98     | Echo (ping) reply id=0xf11, seq=2/512, t     |
| 24   | 202.133872 | 192.168.1.10 | 192.168.1.1  | ICMP     | 98     | Echo (ping) request id=0x1011, seq=3/768, t  |
| 25   | 202.135883 | 192.168.1.1  | 192.168.1.10 | ICMP     | 98     | Echo (ping) reply id=0x1011, seq=3/768, t    |
| 26   | 203.137468 | 192.168.1.10 | 192.168.1.1  | ICMP     | 98     | Echo (ping) request id=0x1111, seq=4/1024, t |
| 27   | 203.138373 | 192.168.1.1  | 192.168.1.10 | ICMP     | 98     | Echo (ping) reply id=0x1111, seq=4/1024, t   |
| → 28 | 204.140321 | 192.168.1.10 | 192.168.1.1  | ICMP     | 98     | Echo (ping) request id=0x1211, seq=5/1280, t |
| ← 29 | 204.141088 | 192.168.1.1  | 192.168.1.10 | ICMP     | 98     | Echo (ping) reply id=0x1211, seq=5/1280, t   |

> Frame 28: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

> Ethernet II, Src: Private\_66:68:00 (00:50:79:66:68:00), Dst: 0c:8c:c6:b4:00:00 (0c:8c:c6:b4:00:00)

▼ Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.1

- 0100 .... = Version: 4
- .... 0101 = Header Length: 20 bytes (5)
- > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)  
Total Length: 84  
Identification: 0x1112 (4370)
- > 000. .... = Flags: 0x0  
...0 0000 0000 0000 = Fragment Offset: 0  
Time to Live: 64  
Protocol: ICMP (1)  
Header Checksum: 0xe63b [validation disabled]  
[Header checksum status: Unverified]  
Source Address: 192.168.1.10  
Destination Address: 192.168.1.1  
[Stream index: 1]

▼ Internet Control Message Protocol

- Type: Echo (ping) request (8)
- Code: 0
- Checksum: 0x0df6 [correct]  
[Checksum Status: Good]
- Identifier (BE): 4625 (0x1211)

|      |       |
|------|-------|
| 0000 | 0c 8c |
| 0010 | 00 54 |
| 0020 | 01 01 |
| 0030 | 0e 0f |
| 0040 | 1e 1f |
| 0050 | 2e 2f |
| 0060 | 3e 3f |

## Выводы по работе

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В ходе выполнения лабораторной работы были смоделированы простейшие сети в GNS3 на базе коммутатора Ethernet и маршрутизаторов FRR и VyOS. Выполнена статическая настройка IP-адресации в подсети 192.168.1.0/24 и проверена связность между узлами с использованием ICMP эхо-запросов.

С помощью Wireshark были захвачены и проанализированы ARP- и ICMP-сообщения, а также особенности формирования трафика при выполнении ping в режимах ICMP/UDP/TCP. Поставленные задачи выполнены в полном объёме.